

**LAPORAN PRAKTIKUM**  
**PENGELOLAAN ADMINISTRASI & TEKNOLOGI INFORMASI**  
**PROJECT-BASED: Mini Server Linux untuk Administrasi Sistem**

Disusun Oleh:

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## 1. Studi Kasus

Sebuah institusi membutuhkan mini server Linux untuk kebutuhan internal. Server harus memenuhi kriteria: stabil saat boot, aman dari kegagalan disk, dapat dimonitor performanya, selalu up-to-date, dan siap digunakan oleh banyak pengguna. Kelompok kami bertugas sebagai tim administrator sistem untuk merancang, mengimplementasikan, dan mendemonstrasikan sistem tersebut.

Seluruh konfigurasi dilakukan menggunakan **Ansible** sebagai alat Infrastructure-as-Code untuk menjamin reproduktibilitas dan konsistensi deployment.

## 2. Arsitektur Sistem

| Komponen             | Detail                                     |
|----------------------|--------------------------------------------|
| Host OS              | Windows 10/11                              |
| Control Node         | WSL (Ubuntu) — menjalankan Ansible         |
| Hypervisor           | Oracle VirtualBox                          |
| Guest OS             | Ubuntu Server 24.04.4 LTS (Headless)       |
| Kernel               | 6.8.0-110-generic (x86_64)                 |
| Boot Loader          | GRUB 2.12                                  |
| Jaringan — NAT       | Akses internet & update paket              |
| Jaringan — Host-Only | 192.168.56.101 (manajemen Ansible & Samba) |
| Disk Primer          | sda — 20 GB (sistem /, /boot)              |
| Disk Sekunder        | sdb — 20 GB → sdb1 /data (ext4)            |
| User                 | adminuser (SSH + Samba)                    |
| Timezone             | Asia/Jakarta                               |

Gambar 1 — Login & System Info Ubuntu Server

```
mini-server login: adminuser
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Apr 24 06:59:41 PM UTC 2026

System load:          0.08
Usage of /:            47.7% of 9.75GB
Memory usage:         13%
Swap usage:           0%
Processes:            129
Users logged in:      1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fec9:63e7

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

adminuser@mini-server:~$ systemctl status multipathd
* multipathd.service - Device-Mapper Multipath Device Controller
   Loaded: loaded (/usr/lib/systemd/system/multipathd.service; disabled; preset: enabled)
   Active: inactive (dead) since Fri 2026-04-24 18:56:52 UTC; 3min 7s ago
   Duration: 2min 49.792s
   TriggeredBy: ● multipathd.socket
   Process: 9465 ExecStart=/sbin/multipathd -d -s (code=exited, status=0/SUCCESS)
   Main PID: 9465 (code=exited, status=0/SUCCESS)
   Status: "shutdown"
   CPU: 406ms

Apr 24 18:54:01 mini-server systemd[1]: Stopped multipathd.service - Device-Mapper Multipath Device Controller.
Apr 24 18:54:01 mini-server systemd[1]: multipathd.service: Consumed 1.155s CPU time, 22.1M memory peak, 0B memory swap peak.
Apr 24 18:54:01 mini-server systemd[1]: Starting multipathd.service - Device-Mapper Multipath Device Controller...
Apr 24 18:54:02 mini-server multipathd[9465]: multipathd v0.9.4: start up
Apr 24 18:54:02 mini-server multipathd[9465]: reconfigure: setting up paths and maps
Apr 24 18:54:02 mini-server systemd[1]: Started multipathd.service - Device-Mapper Multipath Device Controller.
Apr 24 18:56:52 mini-server multipathd[9465]: multipathd: shut down
Apr 24 18:56:52 mini-server systemd[1]: Stopping multipathd.service - Device-Mapper Multipath Device Controller...
Apr 24 18:56:52 mini-server systemd[1]: multipathd.service: Deactivated successfully.
Apr 24 18:56:52 mini-server systemd[1]: Stopped multipathd.service - Device-Mapper Multipath Device Controller.
adminuser@mini-server:~$
```

## 3. Implementasi Teknis

### 3.1 Boot & Service Management

**Boot Loader:** GRUB 2.12 — teridentifikasi melalui *grub-install --version*.

**Kernel:** Linux 6.8.0-110-generic — teridentifikasi melalui *uname -r*.

**Service yang dinonaktifkan:** *multipathd.service* (Device-Mapper Multipath Device Controller). Service ini tidak diperlukan karena VM tidak menggunakan multipath storage.

**Custom Startup Service:** *hello-boot.service* — service sederhana yang mencatat pesan ke journal saat sistem boot, membuktikan kemampuan membuat dan mengaktifkan unit systemd.

Gambar 2 — Status multipathd (disabled)

```
mini-server login: adminuser
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Apr 24 06:59:41 PM UTC 2026

System load:          0.08
Usage of /:           47.7% of 9.75GB
Memory usage:         13%
Swap usage:           0%
Processes:            129
Users logged in:      1
IPV4 address for enp0s3: 10.0.2.15
IPV6 address for enp0s3: fd17:625c:f097:2:a00:27ff:fec9:63e7

Expanded Security Maintenance for Applications is not enabled.

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Enable ESM Apps to receive additional future security updates.
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adminuser@mini-server:~$ systemctl status multipathd
* multipathd.service - Device-Mapper Multipath Device Controller
   Loaded: loaded (/usr/lib/systemd/system/multipathd.service; disabled; preset: enabled)
   Active: inactive (dead) since Fri 2026-04-24 18:56:52 UTC; 3min 7s ago
     Duration: 2min 49.752s
   TriggeredBy: * multipathd.socket
   Process: 9465 ExecStart=/sbin/multipathd -d -s (code=exited, status=0/SUCCESS)
   Main PID: 9465 (code=exited, status=0/SUCCESS)
   Status: "shutdown"
     CPU: 406ms

Apr 24 18:54:01 mini-server systemd[1]: Stopped multipathd.service - Device-Mapper Multipath Device Controller.
Apr 24 18:54:01 mini-server systemd[1]: multipathd.service: Consumed 1.155s CPU time, 22.1M memory peak, 0B memory swap peak.
Apr 24 18:54:01 mini-server systemd[1]: Starting multipathd.service - Device-Mapper Multipath Device Controller...
Apr 24 18:54:02 mini-server multipathd[9465]: multipathd v0.9.4: start up
Apr 24 18:54:02 mini-server multipathd[9465]: reconfigure: setting up paths and maps
Apr 24 18:54:02 mini-server systemd[1]: Started multipathd.service - Device-Mapper Multipath Device Controller.
Apr 24 18:56:52 mini-server multipathd[9465]: multipathd: shut down
Apr 24 18:56:52 mini-server systemd[1]: Stopping multipathd.service - Device-Mapper Multipath Device Controller...
Apr 24 18:56:52 mini-server systemd[1]: multipathd.service: Deactivated successfully.
Apr 24 18:56:52 mini-server systemd[1]: Stopped multipathd.service - Device-Mapper Multipath Device Controller.
adminuser@mini-server:~$
```

Gambar 3 — Status hello-boot.service (enabled, executed)

```
adminuser@mini-server:~$ systemctl status hello-boot.service
* hello-boot.service - Custom Welcome Boot Service
   Loaded: loaded (/etc/systemd/system/hello-boot.service; enabled; preset: enabled)
   Active: inactive (dead) since Fri 2026-04-24 18:54:54 UTC; 6min ago
     Main PID: 11770 (code=exited, status=0/SUCCESS)
       CPU: 8ms

Apr 24 18:54:54 mini-server systemd[1]: Starting hello-boot.service - Custom Welcome Boot Service...
Apr 24 18:54:54 mini-server systemd[1]: hello-boot.service: Deactivated successfully.
Apr 24 18:54:54 mini-server systemd[1]: Finished hello-boot.service - Custom Welcome Boot Service.
adminuser@mini-server:~$ _
```



### 3.3 Keamanan Data (Backup & Recovery)

Kelompok memilih opsi **mekanisme cadangan data manual** menggunakan *rsync* dan *cron*.

#### Konfigurasi Cron:

```
0 2 * * * rsync -a /data/ /backup/ > /var/log/data_backup.log 2>&1
```

Backup otomatis berjalan setiap hari pukul 02:00 WIB.

#### Simulasi Kegagalan & Pemulihan:

1. File *Hi.txt* dibuat di */data/share/* dari PC Windows melalui Samba
2. File dihapus: `sudo rm /data/share/Hi.txt`
3. Restore dari backup: `sudo rsync -a /backup/share/ /data/share/`
4. Verifikasi: `cat /data/share/Hi.txt` — data berhasil dipulihkan

Gambar 5 — Crontab entry untuk backup

```
adminuser@mini-server:~$ sudo crontab -l -u root
#Ansible: Backup Data Partition
0 2 * * * rsync -a /data/ /backup/ > /var/log/data_backup.log 2>&1
adminuser@mini-server:~$
```

Gambar 6 — Simulasi delete & recovery

```
adminuser@mini-server:/data/share$ sudo rsync -a /data/ /backup/
adminuser@mini-server:/data/share$ ls
Hi.txt
adminuser@mini-server:/data/share$ cat Hi.txt
Hello, this is the host PC!
adminuser@mini-server:/data/share$ sudo rm /data/share/Hi.txt
adminuser@mini-server:/data/share$ sudo rsync -a /backup/share/ /data/share/
adminuser@mini-server:/data/share$ cat /data/share/Hi.txt
Hello, this is the host PC!
adminuser@mini-server:/data/share$ _
```

### 3.4 Monitoring Sistem

Tools yang diinstall: **htop**, **sysstat** (sar/iostat), dan **stress**.

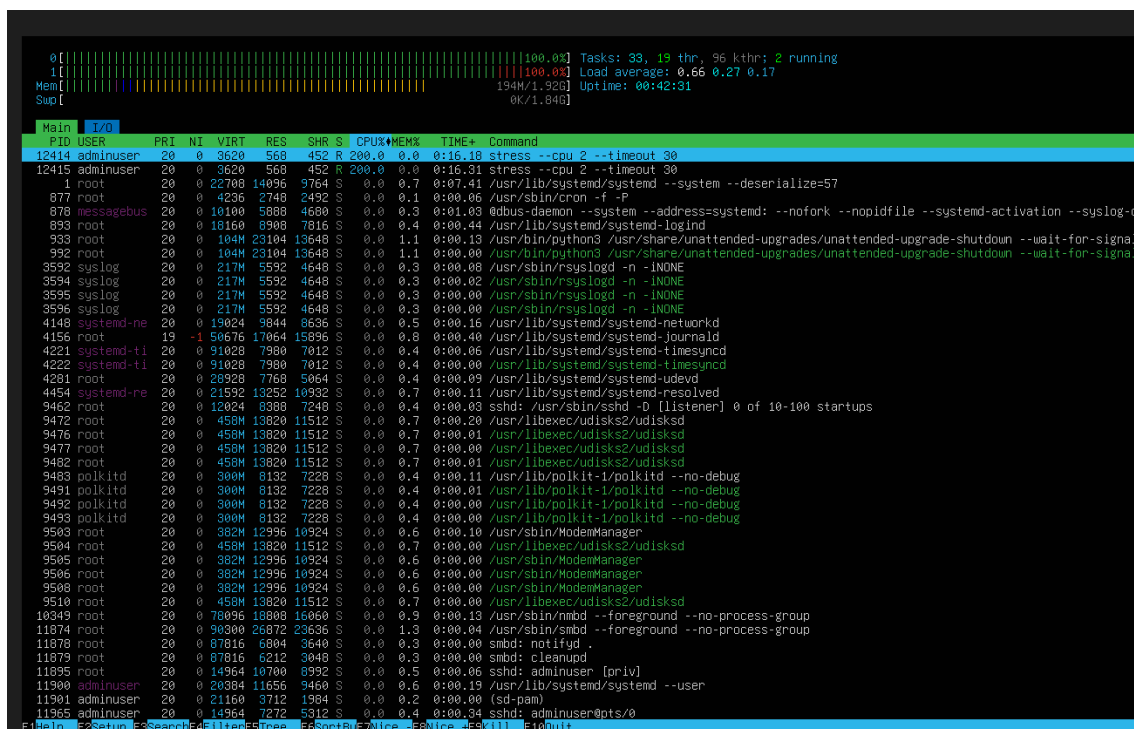
**Simulasi beban tinggi:**

```
stress --cpu 2 --timeout 30
```

Perintah ini membebankan 2 CPU worker selama 30 detik.

**Analisis:** Saat stress test berjalan, kedua core CPU mencapai 100% utilisasi. Proses *stress* mengkonsumsi masing-masing 200% CPU (per-core). Load average naik dari ~0.08 ke ~0.66. Penggunaan RAM tetap stabil di ~194 MB / 1.9 GB karena stress test hanya membebani CPU. Setelah timeout berakhir, CPU kembali normal dalam beberapa detik.

Gambar 7 — htop saat stress test (CPU 100%)



Gambar 8 — htop setelah stress berakhir

```
adminuser@mini-server:~$ systemctl status smbd
● smbd.service - Samba SMB Daemon
   Loaded: loaded (/usr/lib/systemd/system/smbd.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-04-24 18:54:56 UTC; 10min ago
     Docs: man:smbd(8)
           man:samba(7)
           man:smb.conf(5)
  Main PID: 11874 (smbd)
    Status: "smbd: ready to serve connections..."
     Tasks: 3 (limit: 2263)
    Memory: 7.5M (peak: 7.9M)
       CPU: 71ms
    CGroup: /system.slice/smbd.service
            └─11874 /usr/sbin/smbd --foreground --no-process-group
              └─11878 'smbd: notifyd' ,
                └─11879 'smbd: cleanupd' "
```



### 3.5 Update Sistem & Keamanan Dasar

**Update:** Sistem di-update melalui *apt update* && *apt upgrade -y*.

**Firewall (UFW):**

| Port/Service    | Action | Keterangan                            |
|-----------------|--------|---------------------------------------|
| 22/tcp (SSH)    | ALLOW  | Akses remote administrasi             |
| Samba           | ALLOW  | File sharing ke Windows               |
| 23/tcp (Telnet) | DENY   | Ditutup karena tidak aman (plaintext) |

Gambar 9 — apt update

```
adminuser@mini-server:~$ sudo apt update
Hit:1 http://id.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://id.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://id.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
adminuser@mini-server:~$
```

Gambar 10 — ufw status

```
adminuser@mini-server:~$ sudo ufw status
[sudo] password for adminuser:
Status: active

To Action From
--
22/tcp ALLOW Anywhere
Samba ALLOW Anywhere
23/tcp DENY Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
Samba (v6) ALLOW Anywhere (v6)
23/tcp (v6) DENY Anywhere (v6)

adminuser@mini-server:~$ _
```

### 3.6 Layanan Tambahan — Samba File Sharing (Opsi B)

Kelompok memilih **Opsi B: File Sharing** menggunakan Samba.

**Konfigurasi share:** Folder `/data/share` di-share sebagai *PraktikumShare* dengan autentikasi user-level (adminuser).

**Akses dari Windows:** Melalui File Explorer → `\\192.168.56.101`

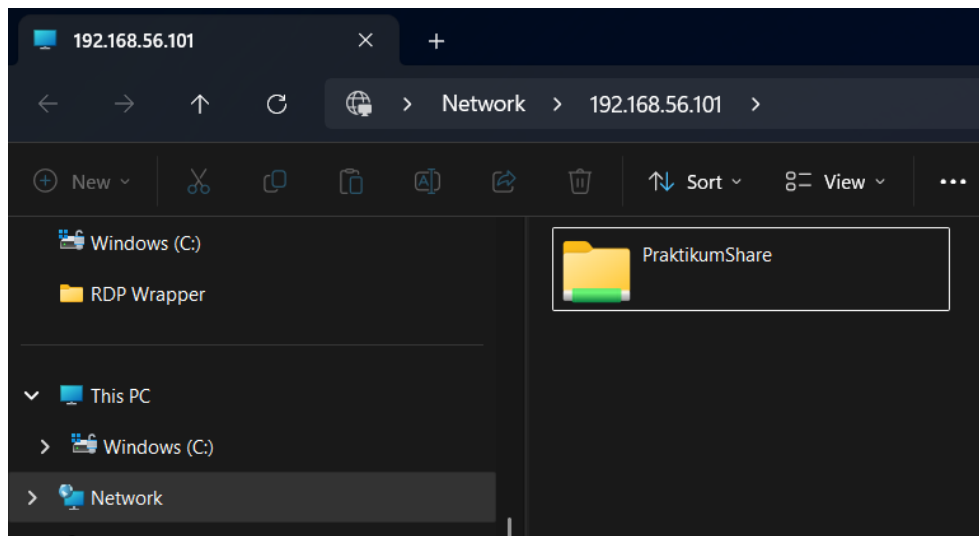
**Pengujian:** File *Hi.txt* berhasil dibuat dari Windows, terbaca di server Linux, dan sebaliknya — membuktikan file sharing dua arah berfungsi.

Gambar 11 — Status Samba (smbd active)

```
adminuser@mini-server:~$ systemctl status smbd
● smbd.service - Samba SMB Daemon
   Loaded: loaded (/usr/lib/systemd/system/smbd.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-04-24 18:54:56 UTC; 10min ago
     Docs: man:smbd(8)
           man:samba(7)
           man:smb.conf(5)
   Main PID: 11874 (smbd)
    Status: "smbd: ready to serve connections..."
     Tasks: 3 (limit: 2263)
    Memory: 7.5M (peak: 7.9M)
       CPU: 71ms
    CGroup: /system.slice/smbd.service
            └─11874 /usr/sbin/smbd --foreground --no-process-group
              └─11878 "smbd: notifyd"
                └─11879 "smbd: cleanupd"

Apr 24 18:54:56 mini-server systemd[1]: Starting smbd.service - Samba SMB Daemon...
Apr 24 18:54:56 mini-server (smbd)[11874]: smbd.service: Referenced but unset environment variable evaluates to an empty string: SMBOPTIONS
Apr 24 18:54:56 mini-server systemd[1]: Started smbd.service - Samba SMB Daemon.
adminuser@mini-server:~$
```

Gambar 12 — Akses dari Windows Explorer



Gambar 13 — Isi folder PraktikumShare

|                                           |      |                   |               |      |
|-------------------------------------------|------|-------------------|---------------|------|
| Network > 192.168.56.101 > PraktikumShare |      |                   |               |      |
| Sort View ...                             |      |                   |               |      |
|                                           | Name | Date modified     | Type          | Size |
|                                           | Hi   | 4/25/2026 2:09 AM | Text Document | 1 KB |

Gambar 14 — Verifikasi isi file dari server

```
adminuser@mini-server:/data/share$ cat Hi.txt
Hello, this is the host PC!adminuser@mini-server:/data/share$ ls
Hi.txt
adminuser@mini-server:/data/share$ _
```

## 4. Masalah & Solusi

| No | Masalah                                                                      | Solusi                                                                                    |
|----|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1  | Timezone VM default UTC, cron tidak sinkron dengan waktu server Asia/Jakarta | Mengubah timezone server Asia/Jakarta: <code>timedatectl set-timezone Asia/Jakarta</code> |
| 2  | Password Samba & Linux terpisah — akses tidak dari Windows                   | Menggunakan <code>winsw</code> -a adminuser untuk sinkronisasi kredensial                 |
| 3  | multipathd aktif tanpa multipath storage (warning aktif)                     | Mengaktifkan service: <code>systemctl disable --now multipathd</code>                     |
| 4  | Backup directory /backup belum ada saat pertama kali boot                    | Membuatkan directory /backup otomatis sebelum konfigurasi cron                            |

## 5. Analisis Administrator

Dari implementasi ini, kami mengambil beberapa pelajaran penting sebagai administrator sistem:

- **Automasi adalah kunci:** Menggunakan Ansible memungkinkan seluruh konfigurasi server direproduksi dalam hitungan menit, mengurangi human error.
- **Pemisahan disk** antara sistem dan data adalah praktik terbaik untuk isolasi kegagalan.
- **Backup terjadwal** dengan rsync + cron memberikan jaring pengaman minimum tanpa memerlukan solusi RAID yang lebih kompleks.
- **Firewall** harus dikonfigurasi secara proaktif — menutup port yang tidak perlu (seperti Telnet/23) mencegah attack surface yang tidak diperlukan.
- **Monitoring** dengan htop memberikan visibilitas real-time yang sangat berguna untuk diagnosis masalah performa.
- **Samba** membuktikan Linux dapat berinteroperasi dengan ekosistem Windows secara seamless untuk kebutuhan file sharing.

## 6. Kesimpulan

Kelompok kami telah berhasil merancang, mengimplementasikan, dan mendemonstrasikan sebuah mini server Linux menggunakan Ubuntu Server 24.04 LTS di lingkungan VirtualBox. Seluruh spesifikasi tugas praktikum telah dipenuhi:

1. Boot loader (GRUB 2.12) dan kernel (6.8.0-110-generic) teridentifikasi, service dikelola melalui systemd.
2. Dua disk dikonfigurasi dengan partisi sistem dan data terpisah.
3. Mekanisme backup rsync + cron diimplementasikan dan recovery berhasil dibuktikan.
4. Monitoring CPU dilakukan dengan htop dan stress test.
5. Sistem di-update dan firewall UFW dikonfigurasi.
6. Samba file sharing berhasil diakses dari Windows.

Seluruh konfigurasi otomatisasi menggunakan Ansible, menjadikan sistem ini sepenuhnya reproducible dan siap untuk deployment ulang kapan saja.